

Question 10

- Data formats refer to the structure that is predefined for encoding. The choice of each depends on the task on hand.

Types of data formats

(a) CSV (Comma separated values)

- This is a text based data format where each line represents a row of data and fields in the row are separated by commas.

- It is chosen for simple and small datasets because of its simplicity and interoperability format.

(b) JSON (JavaScript Object Notation)

- It is also a text based data format that is used for storing and sharing data. It is easy to read and write and is good for storing structured data.

- This format is chosen for more complex data structures than CSV.

(c) Parquet File Format

- This is a columnar data file format and is chosen for large datasets and real time big data processing.

(d) Avro File Format

- This is a binary row-based format. It stores the schema as part of the data file and handles schema changes smoothly. It is chosen for big data environments and for long-running pipelines.